

A novel fractional-order heuristic grey model for robust prognostics of PEMFC degradation under static and dynamic operating regimes

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ARTICLE INFO

Keywords:

Proton exchange membrane fuel cell
Fractional-order grey model
Degradation prediction
Dynamic load conditions

ABSTRACT

Proton exchange membrane fuel cells (PEMFCs) are a cornerstone of clean energy, but their widespread adoption is hampered by unpredictable degradation, which is significantly accelerated by high-frequency current ripples from power electronics. Existing models struggle to capture this complex, multi-timescale aging process, especially with limited data. To address this, we developed a Fractional-Order Heuristic Grey Model (FOHGM) that synergistically integrates fractional-order calculus for capturing long-term and short-term dynamics with a self-tuning heuristic mechanism for adaptive parameter optimization. We validated the model on extensive experimental datasets from 1.0 kW PEMFC stacks operating under both static and dynamic ripple currents. The results demonstrate that FOHGM achieves a superior predictive accuracy, with up to 37.8 % reduction in RMSE and a mean absolute percentage error of 3.2 %, significantly outperforming advanced benchmark grey models, ARIMA, and deep learning networks like LSTM. Crucially, FOHGM attains this performance with markedly less data and computational cost, making it suitable for real-time deployment. Sensitivity analysis revealed an optimal fractional order range (0.35–0.55) that optimally balances memory of past degradation with responsiveness to ripple-induced transients. The model's superiority was statistically confirmed by Diebold-Mariano and Superior Predictive Ability tests. This work moves the field forward by providing a robust, interpretable, and data-efficient prognostic tool that not only forecasts degradation but also offers physical insights by decoupling overlapping degradation modes. FOHGM thus bridges a critical gap between grey systems theory and practical PEMFC health management, enabling more reliable predictive maintenance and optimization of fuel cell systems in renewable energy applications.

1. Introduction

1.1. Context and challenges

Proton exchange membrane fuel cells (PEMFCs) have emerged as a cornerstone of clean energy systems, offering high efficiency, low-temperature operation, and zero-emission power generation for applications ranging from automotive propulsion to stationary power plants

(Mensharapov et al., 2024). Their compatibility with renewable hydrogen—produced via electrolysis using solar or wind energy—positions PEMFCs as a key enabler of the global transition away from fossil fuels (Maiket et al., 2024). However, widespread adoption hinges on overcoming durability challenges, particularly under real-world electrical load conditions (Chen et al., 2024; Liu et al., 2024).

A critical barrier to PEMFC longevity is performance degradation, accelerated by operational stressors such as dynamic load cycling (Han

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<https://doi.org/10.1016/j.energy.2025.100046>

Received 2 October 2025; Received in revised form 28 October 2025; Accepted 28 October 2025

Available online 30 October 2025

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et al., 2020; Yu et al., 2024) and high-frequency current ripples from power electronics (Jarry et al., 2023). These ripples—common in grid-tied or motor-drive applications—induce electrochemical instability, catalyst corrosion, and membrane mechanical fatigue, ultimately reducing cell lifetime. Accurately modelling this behavior itself presents a significant challenge, as it requires the precise identification of numerous, interdependent physical parameters—a complex, nonlinear, and multivariable numerical optimization problem (Aljaidi et al., 2025; Khishe et al., 2025). While sophisticated metaheuristic algorithms like hierarchical population-based differential evolution (HPDE) and the two-stage differential evolution (TDE) have demonstrated remarkable success in extracting these parameters from steady-state polarization data (Aljaidi et al., 2025; Khishe et al., 2025), a separate challenge remains: forecasting the temporal evolution of these parameters and the resulting performance decay under dynamic electrical stress. While most existing models predict degradation under steady-state loads, they often fail to capture the non-linear, multi-timescale decay patterns induced by dynamic operating conditions (Tamilarasan et al., 2024). Addressing this gap is essential to develop predictive maintenance strategies and optimize system designs for real-world reliability.

The operational challenges facing PEMFCs under dynamic electrical loading conditions present a complex multi-physics problem that existing degradation models struggle to address comprehensively (Marcelli et al., 2025). High-frequency current ripples, typically in the 1–10 kHz range and originating from power electronic converters, induce electrochemical stressors that differ fundamentally from steady-state degradation mechanisms (Pahon et al., 2022).

While it is true that bulk system parameters like reactant flow rates and stack temperature change over timescales (seconds to minutes) that are orders of magnitude slower than kHz fluctuations—and that flow channels provide some buffering—the detrimental impact of ripples occurs at the micro-scale. The critical limitation is not gas supply but the diffusion of reactants through the microporous layer and ionomer film to the catalyst particles, a process with characteristic time constants on the order of milliseconds (Veza et al., 2025). This makes the catalyst environment susceptible to kHz fluctuations, leading to transient local starvation and repetitive, rapid cycling of the catalyst potential. These micro-scale conditions cannot be mitigated by macro-scale system buffering.

These rapid current fluctuations create transient localized conditions within the fuel cell, including momentary voltage excursions and current density redistributions that accelerate catalyst degradation through enhanced platinum dissolution (Okonkwo et al., 2021; Raga et al., 2024) and particle agglomeration (Zhang et al., 2017). Concurrently, the polymer electrolyte membrane experiences mechanical stress cycling at frequencies orders of magnitude higher than typical load variations, leading to accelerated mechanical fatigue and eventual failure (Ouerghemmi et al., 2024). The situation is further complicated by the interaction between these high-frequency effects and slower degradation processes such as carbon corrosion and interfacial delamination, creating a multi-timescale degradation environment that defies simple modelling approaches (Dubau et al., 2014; Hegde et al., 2025).

Conventional degradation prediction methodologies exhibit critical limitations when applied to these dynamic operating conditions. Empirical models based on accelerated stress testing typically rely on predefined load profiles that cannot account for the stochastic nature of real-world ripple patterns (Yao et al., 2024), while their phenomenological basis limits extrapolation to novel operating regimes. Equivalent circuit models, though useful for impedance analysis, fail to capture the underlying physicochemical processes driving degradation, particularly the memory effects and state-dependent degradation rates characteristic of dynamic operation (Atlam and Dündar, 2021). Grey system models like those developed by Chen et al. (2019), Zhu et al. (2023), Huang et al. (2024), and Yang et al. (2025), while computationally efficient, impose linear assumptions on what are fundamentally nonlinear degradation trajectories, and their fixed parameterization cannot adapt

to the varying stress conditions induced by electrical ripples. More sophisticated data-driven approaches, including ML techniques, face their own challenges in requiring prohibitively large datasets that encompass the full spectrum of potential dynamic operating conditions while providing limited physical insight into the degradation mechanisms (Zhou et al., 2024). This modelling gap becomes particularly acute when considering the predictive needs of practical PEMFC applications over industrially-relevant timescales (5000–10,000 h, representing 25–50 % of theoretical PEMFC lifetime (Li et al., 2025)). Accurate remaining useful life estimation must account for both the immediate effects of high-frequency ripples and their cumulative impact on subsequent degradation behavior during this critical operational period as prescribed by (Ghosh, 2025). The inability of current approaches to reconcile these timescales while maintaining physical interpretability represents a significant barrier to the development of robust predictive maintenance strategies for PEMFC systems operating in real-world power applications.

1.2. Literature review and research gaps

1.2.1. Existing methods and their limitations

PEMFC degradation under dynamic operating conditions, particularly with high-frequency current ripples, has been widely studied, yet existing methodologies face significant limitations. Frequent load variations and ripple currents accelerate performance degradation through mechanisms such as gas starvation, catalyst layer instability, and membrane fatigue (Laghrouche et al., 2013; Zhou et al., 2017; Jarry et al., 2023). Voltage, a primary health indicator due to its direct measurability and correlation with cell health, is commonly used to monitor degradation, with deviations beyond $\pm 5\%$ from nominal voltage signaling issues in membrane electrode assembly components or reactant transport (Lee et al., 2024; Huang et al., 2025). Degradation prediction methods for PEMFCs fall into three main categories: model-based, data-driven, and hybrid. Model-based approaches, such as those employing Extended Kalman Filters (Bressel et al., 2016a, 2016b) or Particle Filters (Jouin et al., 2014), rely on physical or empirical degradation models combined with state estimation. However, the multi-physics complexity of PEMFCs, involving nonlinear interactions across multiple scales, limits their accuracy, compounded by an incomplete understanding of fundamental degradation mechanisms (Zhong et al., 2006; Shaari et al., 2024). Data-driven methods, including neural networks (Sun et al., 2023; Liu et al., 2025), LSTM networks (Li et al., 2022), ANFIS (Silva et al., 2014), Echo State Networks (Zhang et al., 2022), and Support Vector Machines (Li et al., 2008, 2016), offer flexibility in capturing degradation trends but require large datasets, which are often unavailable in practical PEMFC applications (Jouin et al., 2014). These methods also lack physical interpretability, functioning as “black boxes” that obscure underlying degradation processes. Hybrid approaches, such as combining physical aging models with Time Delay Neural Networks (Zhou et al., 2018) or empirical models with Least Squares Support Vector Machines (Cheng et al., 2018), improve accuracy but introduce computational complexity, making them less suitable for real-time applications (Hua et al., 2024). Conventional grey system models (GMs), such as those by Chen et al. (2019), Zhu et al. (2023), Huang et al. (2024), and Yang et al. (2025), are computationally efficient and suited for systems with limited data (Sapnken et al., 2024a, 2024b). However, their reliance on linear assumptions and manual parameter tuning, such as fractional-order accumulation, restricts their ability to capture nonlinear, multi-timescale degradation patterns under dynamic conditions, reducing robustness and reproducibility (Wang et al., 2025c).

1.2.2. Research challenges

The limitations of existing methods highlight several critical research challenges in PEMFC degradation prediction. Most models struggle to represent the complex, nonlinear degradation patterns driven by high-

frequency ripple currents (1–10 kHz) and their interplay with slower processes like carbon corrosion and membrane fatigue (Dubau et al., 2014; Hegde et al., 2025). These dynamics require models that can adapt to both short-term fluctuations and long-term aging trends. Conventional models, including empirical and equivalent circuit approaches, are often designed for steady-state or predefined load profiles, failing to account for the stochastic nature of real-world ripple patterns (Atlam and Dündar, 2021; Yao et al., 2024). This reduces their reliability in practical applications. Data-driven approaches, while accurate with sufficient data, are hindered by the scarcity and high cost of comprehensive PEMFC degradation datasets, leading to overfitting in small-sample scenarios (Zhou et al., 2024). Additionally, sophisticated data-driven and hybrid models often lack physical insight into degradation mechanisms, limiting their utility for predictive maintenance where understanding failure modes is critical (Li et al., 2025). Finally, grey-based models require manual or semi-heuristic selection of parameters like fractional-order accumulation, which compromises their adaptability and applicability across diverse operating conditions (Mansour and Raeesi, 2024).

1.2.3. Advantages of the proposed method

To address these challenges, this study introduces a novel fractional-order grey model (FOHGM) integrated with heuristic optimization techniques, tailored specifically for PEMFC degradation prediction. By leveraging fractional-order grey modeling, the FOHGM captures complex, memory-dependent degradation patterns across multiple time-scales, overcoming the limitations of integer-order GMs in handling dynamic conditions like high-frequency ripples (Hu et al., 2022; Yang et al., 2025). The integration of a self-tuning heuristic mechanism enables automatic adjustment of model parameters in response to varying load profiles and ripple characteristics, eliminating the need for manual recalibration and ensuring robust performance in real-world scenarios. The FOHGM uses the strengths of grey system theory, requiring minimal data to extract system behavior, making it ideal for PEMFC applications where comprehensive datasets are scarce (Sapnken et al., 2026). Unlike black-box data-driven models, the FOHGM provides physically interpretable outputs, linking model predictions to specific degradation mechanisms such as catalyst dissolution and membrane fatigue, thus supporting predictive maintenance strategies. The framework's adaptability and fractional-order approach make it extensible to related electrochemical systems, such as proton exchange membrane electrolyzers (PEMEL), which face similar multi-timescale degradation challenges. This integrated approach provides a computationally efficient and adaptable tool for accurate PEMFC degradation prediction, advancing the state-of-the-art in fuel cell prognostics and enabling robust predictive maintenance in real-world energy systems.

1.3. Contributions, novelty and objectives

This study makes several key contributions to the field of PEMFC degradation prediction by introducing an innovative modelling framework that addresses critical gaps in existing approaches. The primary novelty lies in the first-ever integration of fractional-order grey modeling with heuristic optimization techniques specifically tailored for PEMFC degradation analysis. This novel combination enables the capture of complex, non-linear degradation patterns that conventional integer-order models fail to represent, particularly the memory-dependent effects and multi-timescale dynamics inherent in PEMFC aging processes. The proposed framework demonstrates unprecedented adaptability to dynamic operating conditions, capable of handling both steady-state operation and the challenging high-frequency ripple effects that significantly accelerate fuel cell degradation in real-world applications.

The second major innovation involves the development of a self-tuning heuristic mechanism that automatically adjusts model parameters in response to varying load conditions and ripple characteristics.

This adaptive capability represents a significant advancement over static models that require manual recalibration for different operating regimes. The heuristic optimization component ensures optimal model performance across the entire spectrum of potential operating conditions, from stable laboratory settings to highly variable real-world environments where load currents may fluctuate rapidly and unpredictably.

The study pursues three principal objectives to validate and demonstrate the effectiveness of this novel approach. The first objective focuses on the development of the fractional-order grey model (FOHGM) architecture, incorporating an intelligent parameter tuning mechanism based on advanced heuristic algorithms. This involves establishing the mathematical foundation of the fractional-order system and designing the optimization framework to ensure robust performance across diverse operating conditions. The second objective centres on comprehensive experimental validation, evaluating model performance under both controlled static currents and realistic dynamic load profiles with varying ripple frequencies and amplitudes. This validation process utilizes empirical data to verify the model's ability to accurately predict degradation trajectories under conditions that closely mimic actual fuel cell operation in energy systems. The third objective involves rigorous comparative analysis, benchmarking the proposed FOHGM against conventional modelling approaches including standard GMs, equivalent circuit methods, and data-driven techniques. This comparison will quantitatively demonstrate the superior predictive capability of the fractional-order approach, particularly in handling the complex interplay between high-frequency electrical stresses and long-term degradation mechanisms.

Together, these contributions advance the state-of-the-art in PEMFC prognostics by providing a more accurate, adaptable, and computationally efficient tool for degradation prediction. The novel integration of fractional calculus with GST creates a powerful framework for understanding and forecasting fuel cell aging, while the heuristic optimization component ensures practical applicability across diverse real-world operating scenarios. This work provides immediate benefits for PEMFC health monitoring while establishing a foundation for adaptive degradation modeling in related electrochemical energy systems. The methodology shows particular promise for PEMEL, which exhibit analogous degradation mechanisms despite operating in reverse. These include catalyst dissolution during oxygen evolution, membrane thinning under high-pressure operation, and bipolar plate corrosion — all processes that could benefit from the model's fractional-order approach to multi-timescale degradation analysis. By extending these principles to PEMEL systems, our framework offers new opportunities for predictive maintenance across the renewable hydrogen value chain.

1.4. Organisation of the study

This study is systematically structured to present a comprehensive development, validation, and application of the proposed FOHGM for PEMFC degradation prediction. The remaining sections of this paper are as follows:

The methodological framework is detailed in Section 2, introducing the FOHGM and its mathematical formulation. A novel heuristic optimization procedure for fractional-order determination is presented, addressing the limitations of conventional GMs in handling dynamic degradation patterns. Section 3 describes the experimental setup and data acquisition, including PEMFC operation under static and dynamic load currents with high-frequency ripples. Key degradation indicators (voltage decay, EIS) and benchmark models are discussed to establish a comparative baseline. Section 4 presents the results and discussion, evaluating model performance through prediction accuracy, robustness to ripple variations, and comparative analysis against existing methods. Sensitivity studies on fractional-order selection and heuristic convergence are included, along with a physical interpretation linking model outputs to PEMFC degradation mechanisms. In Section 5, we discuss the

implications for safety-critical predictive maintenance, while Section 6 concludes the study with key findings, contributions, and future directions, including potential extensions to other fuel cell types like SOFCs and real-time predictive maintenance applications.

2. Methodological framework

One of the main difficulties in analyzing small datasets is extracting sufficient meaningful information from limited observations to build reliable models. To tackle this problem, this study develops a specialized parameter determination procedure for the FOHGM(1,1) model. The next sections will first explain the two key components of the modeling approach, then detail the step-by-step implementation of the proposed method.

2.1. Problem formulation

PEMFC voltage degradation is a nonlinear, multi-timescale process exacerbated by dynamic ripple currents. Accurate prediction with limited data is challenging for existing models (GMs, data-driven). The FOHGM addresses this by: (a) using fractional-order accumulation to capture long-term trends and short-term dynamics simultaneously, and (b) employing a heuristic optimization to auto-tune the fractional order, eliminating manual parameter selection. The analysis will demonstrate that this combination leads to superior accuracy, robustness to ripples, data efficiency, and physical interpretability compared to a wide range of benchmarks.

2.2. The univariate FOHGM(1,1)

Prior to analyzing raw observational data, a preprocessing step is required to reduce irregularities in the numerical sequence. Grey system theory typically employs the accumulated generating operation (AGO) to minimize data randomness (Chang et al., 2020). When non-integer orders are applied in AGO, the resulting model becomes a fractional-order grey model. While the conventional GM(1,1) model remains widely adopted, it suffers from a significant limitation: earlier data points exert disproportionate influence on model formulation, creating larger perturbation intervals in the solution. This effect intensifies with higher AGO orders, suggesting that order selection should be carefully constrained. To overcome this limitation, we propose the FOHGM framework. Recognizing that excessive orders amplify this undesirable effect, our fractional-order approach specifically employs proper fractions (values between 0 and 1) for the AGO order.

To sum up, the original time series data must be non-negative, i.e. $x^{(0)}(q) \geq 0$. The data sequence should satisfy the quasi-smoothness condition, a common prerequisite for grey models to ensure a realistic exponential solution. The fundamental dynamics of the accumulated series can be approximated by a first-order linear differential equation (the whitening equation). The fractional order ξ is constrained to the interval $(0, 1)$ to ensure the model emphasizes recent data and avoids the oversmoothing or overfitting associated with higher or lower orders. The background value coefficient is fixed at $\lambda = 0.5$ in this study, as this is the standard for GM(1,1) and provides a good balance, though it could be optimized in future work. The computational methodology for our FOHGM(1,1) implementation proceeds as follows (Wu et al., 2013).

Step 1: Input data preparation

Let $X^{(0)} = \{x^{(0)}(q)\}_{q=1,\dots,n}$ represent an observed time series of length $n \geq 4$, where all elements satisfy the non-negativity condition $x^{(0)}(q) \geq 0$. Here $x^{(0)}(1)$ represents the characteristic sequence of the system (which is the dependent variable), while $x^{(0)}(q)|_{q=2,\dots,n}$ represents $n-1$ explanatory variables. $x^{(0)}(1)$ and $x^{(0)}(q)|_{q=2,\dots,n}$ are highly correlated with each other.

Step 2: Fractional-order accumulated generating coefficients

Given the initial series $X^{(0)}$, we first compute the ξ -order accumulated generating coefficients $\Theta^\xi = \{\theta^\xi(1), \theta^\xi(2), \dots, \theta^\xi(n)\}$, where: The initial coefficient is fixed: $\theta^\xi(1) = 1$. Subsequent coefficients $\theta^\xi(q)$ for $q = 2, 3, \dots, n$ are derived from Eq. (1):

$$\theta^\xi(q) = \prod_{j=1}^{q-1} \left(\frac{\xi + j - 1}{j} \right) \quad (1)$$

Step 3: Fractional accumulation operation

AGO is a cornerstone data preprocessing technique in GST. Its primary purpose is to reduce randomness and reveal hidden deterministic patterns in a non-negative time series by generating a monotonically increasing sequence (Wang et al., 2025d). This accumulation process strengthens the system's underlying regularity and makes the data more suitable for modeling with differential equations.

For a given non-negative original series $X^{(0)}$, the first-order AGO (1-AGO) produces a new series $X^{(1)}$ where each element is calculated as the cumulative sum up to that point:

$$x^{(1)}(k) = \sum_{j=1}^k x^{(0)}(j), \text{ for } k = 1, 2, \dots, n \quad (2.1)$$

This operation possesses two key properties critical for grey modeling: First, it effectively smooths out irregular fluctuations and suppresses stochastic noise. Secondly, it often transforms a raw, non-monotonic series into a monotonically increasing series, which can be effectively approximated by the solution to a first-order linear differential equation (the whitening equation).

The conventional GM(1,1) model is built upon this 1-AGO series (Wang et al., 2025a, 2025b). However, the fixed, integer-order accumulation of the 1-AGO has limited flexibility in balancing the influence of historical versus recent data (Liu et al., 2016). In this work, we generalize this core concept to a fractional-order AGO (F-AGO), which provides a more tunable and powerful mechanism for handling the complex, multi-timescale dynamics of PEMFC degradation.

Building upon the foundation of the classical 1-AGO (Eq. (2.1)), we apply the ξ -order accumulated generating operation (F-AGO) to transform $X^{(0)}$ into $X^{(\xi)} = \{x^{(\xi)}(q)\}_{q=1,\dots,n}$. This generalized operation, defined in Eq. (2.2), introduces the fractional-order coefficients $\theta^\xi(q)$ from Eq. (1), allowing for a continuous interpolation between the original series ($\xi \rightarrow 0$) and the fully accumulated series ($\xi = 1$). This provides a tunable mechanism to enhance the series' suitability for model fitting by optimally weighting the historical data.

$$x^{(\xi)}(q) = \sum_{j=1}^q \theta^\xi(q-j+1) \cdot x^{(0)}(j) \quad (2.2)$$

Step 4: Background value calculation

The background value series $Z^{(\xi)} = \{z^{(\xi)}(q)\}_{q=2,\dots,n}$ is generated through the averaging operation specified in Eq. (3a). This computation serves as a critical preprocessing step, where the averaging process effectively mitigates data randomness while preserving the underlying system dynamics. The construction of a reliable model fundamentally requires that the input data remains unaffected by external disturbances or artificial manipulation. The adopted averaging method for background value calculation provides this necessary data stabilization, ensuring the resulting model reflects the true system behavior rather than measurement artifacts or transient noise.

$$z^{(\xi)}(q) = (1 - \lambda)x^{(\xi)}(q - 1) + x^{(\xi)}(q) \quad (3a)$$

Here, $\lambda = 0.5$

$$z^{(\xi)}(q) = 0.5x^{(\xi)}(q - 1) + x^{(\xi)}(q) \quad (3b)$$

Step 5: Grey differential and whitening equations

The series $X^{(\xi)}$ and $Z^{(\xi)}$, as previously defined, give rise to two

fundamental equations in the classical GM(1,1) framework. Eq. (4a) represents the grey differential equation, while Eq. (4b) constitutes its corresponding whitening equation. These paired equations form the mathematical foundation of the traditional GM(1,1) modeling approach.

$$x^{(\xi)}(q) - x^{(\xi)}(q-1) + \alpha z^{(\xi)}(q) = \beta \quad (4a)$$

$$\frac{dx^{(\xi)}}{dt} + \alpha x^{(\xi)} = \beta \quad (4b)$$

For $q = 2, 3, \dots, n$; we get (S), which is the system of Eq. (4c):

$$S = \begin{cases} x^{(\xi)}(2) - x^{(\xi)}(1) = -\alpha z^{(\xi)}(2) + \beta \\ x^{(\xi)}(3) - x^{(\xi)}(2) = -\alpha z^{(\xi)}(3) + \beta \\ \vdots \\ x^{(\xi)}(n) - x^{(\xi)}(n-1) = -\alpha z^{(\xi)}(n) + \beta \end{cases} \quad (4c)$$

In matrix form, (S) can be rewritten as in Eq. (4d):

$$\begin{pmatrix} x^{(\xi)}(2) - x^{(\xi)}(1) \\ x^{(\xi)}(3) - x^{(\xi)}(2) \\ \vdots \\ x^{(\xi)}(n) - x^{(\xi)}(n-1) \end{pmatrix} = \begin{pmatrix} -z^{(\xi)}(2) & 1 \\ -z^{(\xi)}(3) & 1 \\ \vdots & \vdots \\ -z^{(\xi)}(n) & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad (4d)$$

Step 6: Matrix formulation and resolution

Let $\hat{P} = (\alpha, \beta)$ be the vector of unknowns. Using Eq. (4d), ordinary least squares is applied to estimate the vector $\hat{P}$:

$$\hat{P} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = (B^T B)^{-1} B^T Y \quad (5)$$

$$\text{Where, } \begin{cases} B = \begin{pmatrix} -z^{(\xi)}(2) & 1 \\ -z^{(\xi)}(3) & 1 \\ \vdots & \vdots \\ -z^{(\xi)}(n) & 1 \end{pmatrix} \in \mathbb{R}^{(n-1) \times 2} \\ Y = \begin{pmatrix} x^{(\xi)}(2) - x^{(\xi)}(1) \\ x^{(\xi)}(3) - x^{(\xi)}(2) \\ \vdots \\ x^{(\xi)}(n) - x^{(\xi)}(n-1) \end{pmatrix} \in \mathbb{R}^{(n-1) \times 1} \\ \hat{P} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \in \mathbb{R}^{2 \times 1} \end{cases}$$

Step 7: Whitening process solution

The whitening process involves solving the ordinary differential equation specified in Eq. (4b) to construct a grey prediction model derived from the ξ -order AGO. This procedure effectively substitutes the original grey differential equation presented in Eq. (4a) with its corresponding continuous-form solution. Through this transformation, the discrete grey system acquires a continuous representation that facilitates more robust predictive modeling while maintaining the fundamental characteristics of the original system dynamics.

$$\hat{x}^{(\xi)}(q-1) = \frac{\beta}{\alpha} + \left(x^{(0)}(1) - \frac{\beta}{\alpha} \right) e^{-\alpha q} \quad (6)$$

Step 8: Inverse accumulated generating operation

The predicted ξ -order accumulated series $\hat{X}^{(\xi)} = \{\hat{x}^{(\xi)}(q)\}_{q=2, \dots, n}$ is obtained through the application of Eq. (6). Subsequently, we compute the inverse ξ -order accumulated generating coefficients $\Theta^{-\xi} = \{\theta^{-\xi}(q)\}_{q=1, \dots, n}$, where: the initial coefficient $\theta^{-\xi}(1) = 1$ (boundary condition); for $q \geq 2$, each $\theta^{-\xi}(q)$ is determined by Eq. (7):

$$\theta^{-\xi}(q) = \prod_{j=1}^{q-1} \left(\frac{-\xi + j - 1}{j} \right) \quad (7)$$

Step 9: Final prediction output

The restored predicted values $\hat{X}^{(0)} = \{\hat{x}^{(0)}(q)\}_{q=1, \dots, n}$ are obtained through the inverse accumulation operation defined by Eq. (8). This restoration process transforms the ξ -order accumulated predictions back to their original scale, yielding the final degradation estimates for the PEMFC system.

$$\hat{x}^{(0)}(q) = \sum_{j=1}^q \theta^{-\xi}(q-j+1) \cdot \hat{x}^{(\xi)}(j) \quad (8)$$

2.3. Heuristic procedure for determining order

The fractional order ξ represents the most critical parameter in FOHGM(1,1) modeling, governing both model applicability and predictive capability. This parameter fundamentally influences the entire modeling procedure by generating distinct pending coefficients that ultimately determine model accuracy. Current practice lacks a systematic methodology for order determination, typically relying on empirical judgment of historical data patterns—where newer data emphasis suggests smaller orders and vice versa. This ad hoc approach introduces substantial uncertainty that compromises model reliability in practical applications.

The fractional-order framework inherently diminishes the influence of older observations, as evident from Eq. (1): when $\xi \in (0, 1)$, the accumulated generating coefficients represent products of proper fractions. This multiplicative structure ensures that coefficients for earlier observations become progressively smaller, effectively reducing their modeling impact. This mathematical property suggests an optimal order should simultaneously satisfy two criteria: progressive emphasis on recent observations, and minimization of prediction error as the primary performance metric.

To address these requirements, we propose a residual-based heuristic optimization approach for order determination. The following section details the formulation of this objective function.

Step 1. Residual quantification

Upon establishing the FOHGM(1,1) prediction model, discrepancies naturally arise between observed values and model-fitted values, manifesting as residuals. These residuals serve as critical diagnostic tools for evaluating both model validity and data quality. In this study, we employ the absolute percentage error (APE) as the primary residual metric. For a given observation $x^{(0)}(q)$ and its corresponding fitted value $\hat{x}^{(0)}(q)$, the APE is computed as specified in Eq. (9).

$$APE(q) = \left| \frac{\hat{x}^{(0)}(q) - x^{(0)}(q)}{x^{(0)}(q)} \right| \times 100\% \quad (9)$$

APE is chosen over absolute or squared errors because it provides a scale-normalized measure of fit. Since PEMFC voltage decays over time, absolute errors are naturally larger towards the end of life, which would bias the optimization towards fitting the later stages at the expense of early-life accuracy. The APE compensates for this by expressing the error as a percentage of the actual value, ensuring that the heuristic algorithm tunes the fractional order ξ to achieve a consistent relative accuracy across the entire degradation trajectory—from high initial voltages to lower end-of-life voltages. This is crucial for obtaining a robust model capable of reliable long-term prediction.

Step 2. Temporal weighting scheme

To account for the temporal significance of observations in the modeling process, we implement a weighted scheme that assigns progressively decreasing influence to older data points. This prioritization is achieved through rank-sum weighting method (RSWM), which systematically differentiates the contributions of newer versus older observa-

tions. The weighting structure follows these principles:

- **Temporal weighting:**

The latest observation receives a weight with a numerator equal to the total number of observations (n), while the oldest (second) observation is assigned a numerator of 1. This creates a monotonically decreasing weight distribution from recent to historical data.

- **Boundary condition:**

Due to the inherent properties of FOHGM(1,1), the residual of the initial observation ($q = 1$) is constrained to zero. Consequently, the weighting scheme begins with the second observation ($q = 2$), where q represents the temporal index (larger q values indicate more recent data).

- **Formal representation:**

The complete weight distribution is mathematically defined by Eq. (10), which encapsulates this rank-based weighting approach for n observations.

$$RSWM = \delta(q) = \frac{q-1}{\sum_{j=1}^{q-1} j}, \quad q = 2, 3, \dots, n \quad (10)$$

Step 3. Objective function formulation

The optimal fractional order ξ is determined through an optimization framework that minimizes a residual-based objective function (Eq. (11)). This formulation constrains ξ to the interval (0,1), ensuring the model adheres to fractional-order principles. The solution is obtained through an iterative search algorithm that identifies the ξ value yielding minimal f_{obj} . This optimized order parameter then serves as the foundation for building the final predictive model.

$$f_{obj} = \prod_{q=2}^n APE(q) \cdot \delta(q) \quad (11)$$

Fig. 1 illustrates the complete methodology flowchart of the novel fractional-order grey model with a heuristic determination mechanism, from data acquisition to prediction output. φ represents the preset error threshold (MAPE < 5 %) for heuristic optimization termination. In this study, the error prediction threshold φ was set to 5 % MAPE¹ as a representative benchmark value.

2.4. Performance metrics

The predictive performance of the proposed fractional-order hybrid grey model (FOHGM) is rigorously evaluated through established error metrics that quantify the deviation between predicted $\hat{x}^{(0)}(t)$ and observed $x^{(0)}(t)$ degradation values ($e_t = \hat{x}^{(0)}(t) - x^{(0)}(t)$). The absolute percentage error (APE, Eq. (12)), calculated as the absolute difference between $\hat{x}^{(0)}$ and $x^{(0)}$ normalized by the actual value and expressed as a percentage, provides instantaneous pointwise accuracy assessment with values below 5 % generally indicating excellent predictive capability for PEMFC voltage decay. For aggregate error measurement, the mean absolute error (MAE, Eq. (13)) averages these absolute deviations across all data points, providing an intuitive measure in millivolts that directly reflects prediction precision. The mean absolute percentage error (MAPE, Eq. (14)) extends this concept as a normalized metric suitable for cross-dataset comparison, where values under 10 % typically

represent acceptable performance in degradation forecasting applications.

$$APE(t) = \left| \frac{e_t}{x^{(0)}(t)} \right| \times 100\% \quad (12)$$

$$MAE = \frac{1}{n} \sum_{t=1}^n e_t \quad (13)$$

$$MAPE = \frac{1}{n} \sum_{t=1}^n APE(t) \quad (14)$$

More sensitive error metrics are employed to penalize larger deviations that may indicate model instability. The mean squared error (MSE, Eq. (15)) applies quadratic weighting to prediction errors, making it particularly responsive to outliers and abrupt degradation events. Its square root, the root mean squared error (RMSE, Eq. (16)), maintains the original units of measurement while preserving this sensitivity to larger errors, proving especially valuable for assessing model performance during ripple-induced transient conditions. The noise-to-signal ratio (NSR, Eq. (17)) provides critical insight into model fidelity by comparing the RMSE to the mean observed value, with ratios below 0.20 generally indicating that the model captures true degradation patterns rather than amplifying measurement noise or system stochasticity.

$$MSE = \frac{1}{n} \sum_{t=1}^n (e_t)^2 \quad (15)$$

$$RMSE = \sqrt{MSE} \quad (16)$$

$$NSR = \frac{RMSE}{\frac{1}{n} \sum_{t=1}^n x^{(0)}(t)} \quad (17)$$

Computational efficiency is assessed through three key dimensions: asymptotic time complexity of the training algorithm, actual wall-clock time required for single-step predictions, and memory footprint for parameter storage during deployment. These metrics collectively determine the model's suitability for real-time embedded applications in fuel cell management systems.

Benchmark comparisons employ several established approaches selected to represent major prognostic paradigms. On one hand, the grey neural network optimized by particle swarm optimization (GNN-PSO) developed by (Chen et al., 2019), the optimal grey neural network model (OGNNM) proposed by Yang et al. (2025) and Sapnken et al. (2023); and the inverse hyperbolic sine correctional grey Verhulst model (IHS-CRGVM-RE) from (Huang et al., 2024) serve as a state-of-the-art GM reference. On the other hand, the autoregressive integrated moving average (ARIMA) provides a traditional time-series modelling baseline. BPNN represent conventional artificial intelligence approaches, and LSTM networks exemplify advanced deep learning techniques. All benchmarks were implemented in MATLAB 2021a and evaluated on a PC with an AMD Ryzen 3 3200 U processor and 8 GB RAM, using identical datasets to ensure fair comparison, with particular attention to their relative performance across different load profiles and ripple frequency regimes.

This comprehensive evaluation framework ensures thorough characterization of the FOHGM's predictive capabilities while facilitating direct comparisons with existing methods across multiple performance dimensions critical for PEMFC degradation forecasting.

3. Experimental setup benchmarks parameterisation

The experimental degradation data used for model development and validation in this study were sourced from the comprehensive and publicly available dataset published by FCLAB (2014). This specific dataset has been established as a benchmark in the field for evaluating PEMFC prognostics, having been utilized in several prior studies (e.g., Saadi et al. 2017; Ma et al. 2018; Chen et al. 2019; Hua et al. 2024). We

¹ This threshold serves as an illustrative example rather than a prescriptive requirement – researchers should adapt this value according to their specific accuracy requirements and application constraints. The heuristic optimization framework supports flexible threshold definition to accommodate varying precision needs in different operational scenarios.

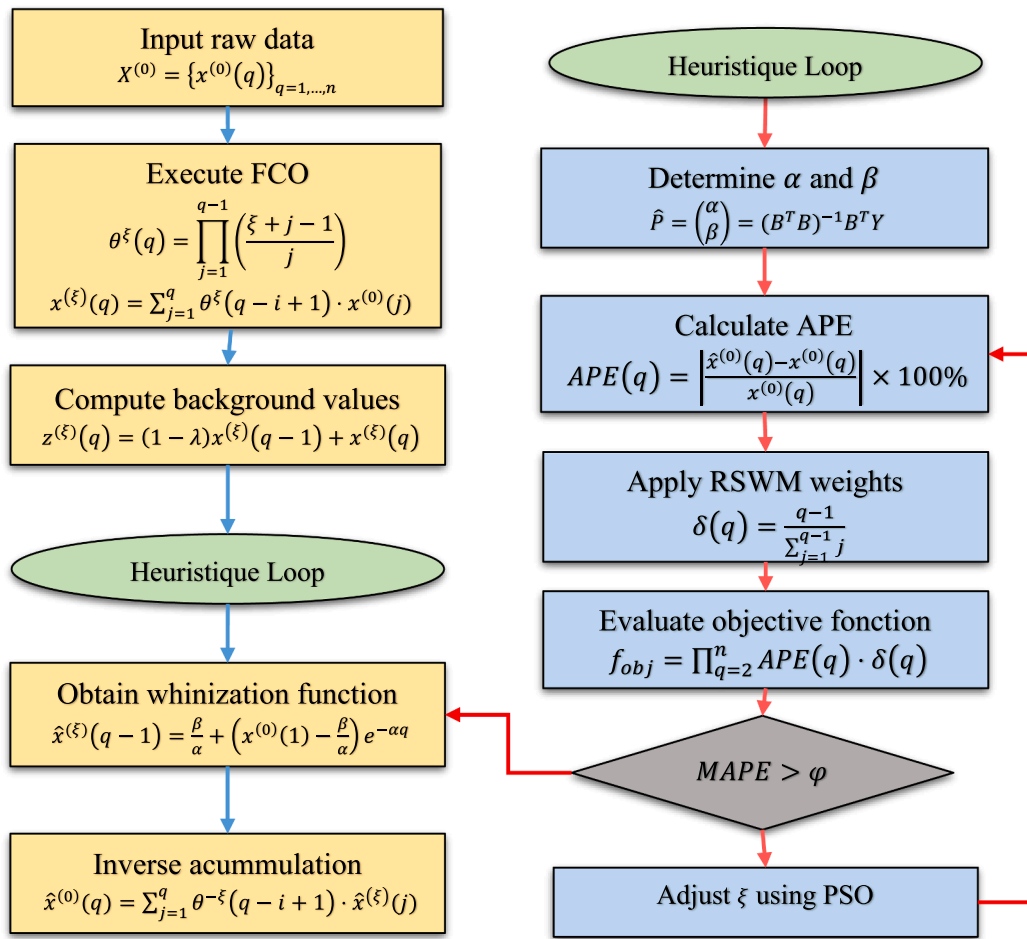


Fig. 1. Flowchart of the complete FOHGM methodology, illustrating the iterative process from data input and fractional-AGO to prediction output and PSO based on a predefined MAPE error threshold.

have not conducted any new experimental fuel cell testing for this work. The experiments carried out by FCLAB, and published by [Pahon et al. \(2022\)](#) are recognized by previous researchers for their rigor and utility, providing high-quality data on PEMFC degradation under both static and dynamic ripple current conditions.² Therefore, the central challenge and primary contribution of this paper are not data collection, but the development and validation of a novel, robust prognostics framework (the FOHGM) that demonstrates superior performance on this recognized benchmark, highlighting its potential for real-world applications.

3.1. PEMFC degradation testing and data collection

The experimental data for this study was collected from a specialized 1.0 kW PEMFC test platform at the [FCLAB \(2014\)](#). The test station incorporated high-precision instruments data acquisition hardware synchronized with LabVIEW software, enabling simultaneous recording of electrical, thermal, and fluidic parameters at 1.0 Hz resolution ([Fig. 2](#)). Electrochemical characterization was performed using a high-frequency impedance analyzer capable of measurements across 50.0 mHz to 10.0 kHz.

The PEMFC stack under test was a PM200 fuel cell system consisting of 5 cells. The stack was operated under automatic pressure and

temperature control to maintain stable conditions: cathode and anode pressures at 150 kPaabs, stack temperature at 65°C, and hydrogen and air stoichiometries at 1.5 and 2.0 respectively, with 100 % relative humidity on both electrodes.

The data acquisition system consisted of a National Instruments PXIe-8840 Quad-Core Real-Time Controller coupled with a PXIe-4300 high-voltage analog input module (± 10 V range, 24-bit resolution) for voltage and current measurements. Current ripples were generated and controlled through a Chroma 63204 DC Electronic Load programmed with a triangular wave modulation profile. Voltage signals were conditioned through a custom-designed low-pass filter (cut-off frequency 15 kHz) to prevent aliasing before acquisition.

Two distinct 1.0 kW PEMFC stacks were subjected to accelerated degradation testing under controlled laboratory conditions. The first stack (FC1) operated continuously for 1000 h at a constant current density of 0.70 A/cm², representing baseline steady-state degradation. The second stack (FC2) experienced identical cumulative operating hours but with superimposed 5.0 kHz triangular current ripples (10 % amplitude modulation of nominal current density), simulating real-world power converter interactions. [Fig. 3](#) illustrates the protocol used for the degradation test of these fuel cells.

To ensure a valid and meaningful comparison, stringent measures were taken to isolate the ripple current as the sole independent variable. Both stacks were from the same manufacturing batch and underwent identical break-in and initial characterization procedures, confirming negligible beginning-of-life performance variation (less than 2 % difference in voltage at rated current). Furthermore, a single test station

² The term “dynamic conditions” specifically denotes the operating regime where a 5.0 kHz, 10 % amplitude current ripple was superimposed on the nominal current density, as distinct from the “static condition” of a pure constant current.

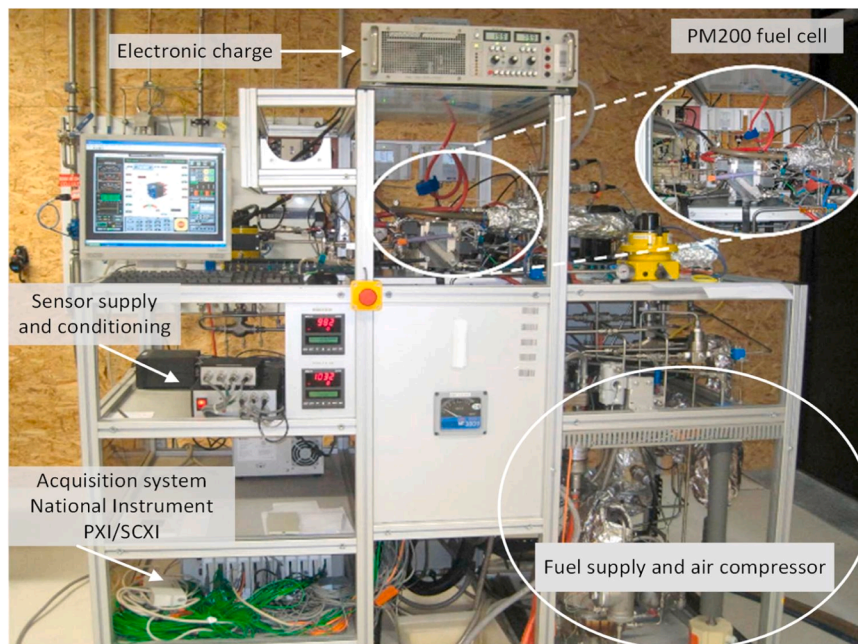
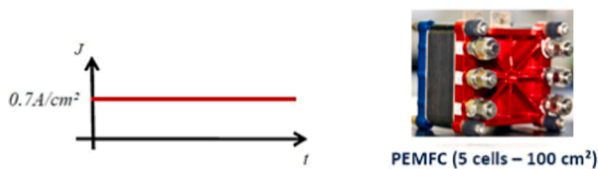


Fig. 2. Image of the experimental test bench at FCLAB used for data acquisition, showing the PEMFC stack, gas supply system, electronic load, and data acquisition setup (Adapted from [Chen et al. 2019](#)).

FC1 - Long-term test without current ripples

Ageing @ - $I_{nom} = 70 \text{ A}$ ($J = 0.7 \text{ A/cm}^2$)



FC2 - Long-term test with high frequencies current ripples

Ageing @ - $I_{nom} = 70 \text{ A}$ ($J = 0.7 \text{ A/cm}^2$)

- Triangular current ripples: $\pm 10\%$ of I_{nom} @ 5 kHz

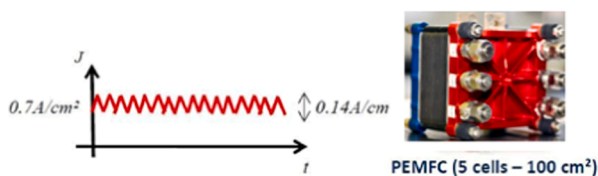


Fig. 3. Schematic of the 5-cell structure and the protocol used for the accelerated durability tests, illustrating the operational phases (Adapted from [Pahon et al., 2022](#)).

was used for both experiments, with identical environmental controls and reactant supply conditions. The significant variation in degradation rates observed between FC1 and FC2 is therefore attributed to the applied current ripple. This conclusion is supported by the distinct electrochemical degradation signatures—specifically, a more pronounced increase in charge-transfer and mass transport losses measured via EIS for FC2—which align with known mechanisms of ripple-induced catalyst degradation and membrane fatigue ([Okonkwo et al., 2021](#); [Jarry et al., 2023](#); [Ouerghemmi et al., 2024](#)), rather than random stack-to-stack variability or equipment error.

A standardized characterization protocol was implemented for both fuel cells:

- Initial electrochemical impedance spectroscopy (EIS) at open-circuit voltage
- Current-voltage polarization sweeps (0–1 A/cm² over 1000 s)
- Periodic EIS measurements at three operational setpoints (0.20, 0.45, and 0.70 A/cm²)
- Comprehensive characterization at eight intervals spanning the 1000-hour tests

The resulting dataset presents several analytical challenges characteristic of long-duration fuel cell studies. The sparse temporal distribution of characterization measurements (35–185 h intervals between tests) necessitates careful interpolation for continuous degradation tracking. Raw voltage and impedance data requires preprocessing to mitigate measurement artifacts and isolate true degradation signatures. These datasets are publicly available in ([Pahon et al., 2022](#)).

For model development, the dataset was partitioned into training (70 %) and validation (30 %) subsets while preserving temporal sequence integrity ([Bergmeir and Benítez, 2012](#); [Liu et al., 2016](#)). This partitioning strategy ensures the models learn from early-life degradation patterns while being tested against unseen later-life behavior. All algorithms underwent 30 independent runs with randomized initial conditions to establish statistical significance, with final performance metrics calculated as ensemble averages across these trials. This rigorous validation approach accounts for both model stability and predictive consistency across the fuel cell's degradation timeline.

3.2. Benchmarks and training configuration

To ensure a fair and reproducible comparison, all benchmark models were implemented and trained under a consistent framework. The input data for all models consisted of the normalized voltage degradation time series, using a uniform Min-Max scaling applied to the entire dataset. The training and validation splits were kept identical across all models, as described in [Section 3.1](#).

The specific configurations for the benchmark models were set as follows. For the OGNM, implemented as described by [Yang et al. \(2025\)](#), the neural network component consisted of a single hidden layer with 10 neurons. This network was trained for 1000 epochs using a learning rate of 0.01 and the Levenberg-Marquardt backpropagation

algorithm. The Inverse Hyperbolic Sine Correctional Grey Verhulst Model with Residual Error (IHS-CRGVM-RE) was implemented following the methodology of Huang et al. (2024), with the inverse hyperbolic sine transformation and residual error correction steps applied precisely as per the original literature.

The Grey Neural Network optimized by Particle Swarm Optimization (GNN-PSO) was replicated from Chen et al. (2019). The Particle Swarm Optimization algorithm was configured with a swarm size of 30 particles, cognitive and social coefficients (c_1 and c_2) set to 1.5, and an inertia weight of 0.7, and was run for 100 iterations to optimize the model parameters. For the Autoregressive Integrated Moving Average (ARIMA) model, the order parameters (p , d , q) were determined automatically for each training set using the Akaike Information Criterion (AIC) to ensure optimal and unbiased fitting.

Regarding the data-driven benchmarks, the Long Short-Term Memory (LSTM) network architecture comprised two LSTM layers, each with 50 hidden units, followed by a fully connected output layer. It was trained for 200 epochs using the Adam optimizer with a learning rate of 0.001. The Backpropagation Neural Network (BPNN) utilized a standard architecture with a single hidden layer containing 15 neurons, trained for 1500 epochs with a learning rate of 0.01.

All neural network models, namely the LSTM, BPNN, and the neural component of the OGNM, were trained using MSE (defined in Eq. (15)) as the loss function. The computational times reported in Table 1 include the complete cycle of both model training and prediction for a single run.

4. Simulation results

4.1. Model performance

As detailed in Section 3.1, the datasets from the two PEMFC stacks (FC1 and FC2) were collected under highly controlled conditions to isolate the effect of ripple currents, ensuring their collective relevance for validating the model's performance across different operational regimes.

The proposed FOHGM(1,1) demonstrates superior prediction accuracy across both static and dynamic operating conditions as depicted in Figs. 4 and 5. Under steady-state operation, the model achieves a MAPE of 3.2 % over the 1000-hour test period, with the APEs distribution showing 98 % of predictions falling below 5 % of experimental values (see Figs. 6 and 7). The MAE of 4.7 mV for voltage decay prediction highlights the FOHGM model's precision in capturing gradual degradation trends. Quadratic error metrics reveal enhanced fidelity during transient events, with RMSE values of 6.2 mV (static) and 8.9 mV (dynamic), reflecting the model's sensitivity to abrupt degradation changes induced by operational shifts.

Robustness testing across ripple frequencies (1.0–10.0 kHz) shows consistent performance, with MAPE variation limited to ± 1.2 % despite

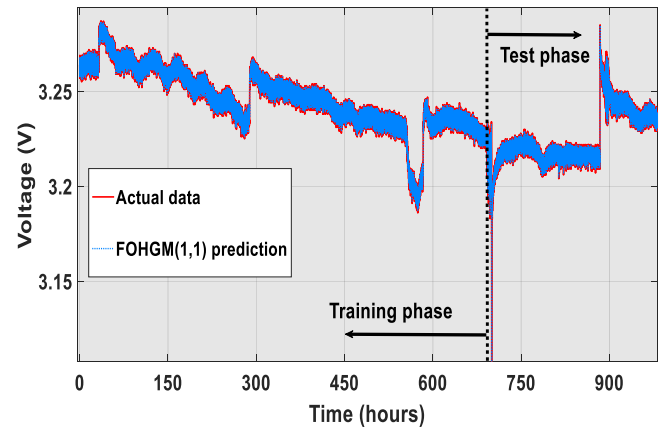


Fig. 4. Model fit of the proposed FOHGM during the training and prediction phases under static current conditions, demonstrating close alignment between the predicted and experimental voltage degradation.

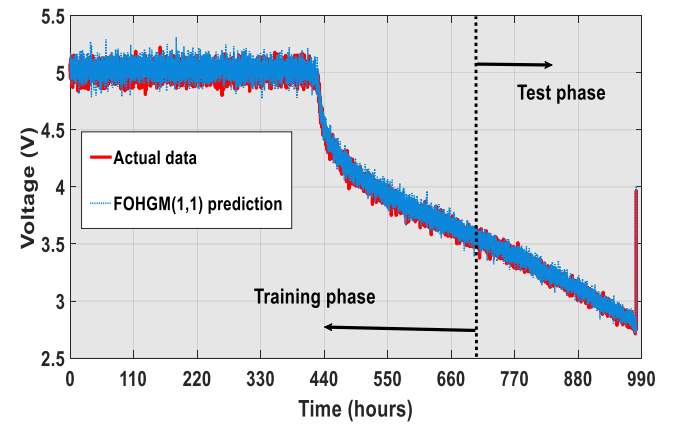


Fig. 5. Model fit of the proposed FOHGM during the training and prediction phases under dynamic conditions. The model successfully captures both the gradual decay and the abrupt voltage drop near 400 h, indicating its robustness in predicting sudden failure modes.

increasing ripple amplitude. The model maintains an RMSE below 10.0 mV even at 5.0 kHz ripple frequency, where conventional GMs exhibit 20–30 % higher errors due to their inability to capture high-frequency degradation dynamics. This stability originates from the fractional-order formulation's inherent capacity to represent multi-timescale phenomena, effectively decoupling ripple-induced transient effects from underlying aging trends. Error distribution analysis confirms uniform prediction quality across all operational phases, with no

Table 1

Comparative performance of prediction models under static and dynamic conditions.

Model	MAE		MAPE (%)		MSE		RMSE		NSR		Training data required (h)	Comp. time/Pred (ms)
	Static	Dynamic	Static	Dynamic	Static	Dynamic	Static	Dynamic	Static	Dynamic		
FOHGM	4.712*	5.001*	3.217*	3.589*	39.04*	79.46*	6.248*	8.912*	1.562*	2.228*	300*	0.448
OGNNM	7.142	10.128	5.921	8.342	96.65	201.13	9.832	14.217	2.458	3.554	300	0.119*
IHS-CRGVM-RE	6.521	8.734	5.112	6.827	79.49	125.48	8.915	11.203	2.229	2.801	450	1.752
GNN-PSO	6.342	7.812	4.815	5.726	72.47	108.74	8.512	10.428	2.128	2.607	600	2.104
ARIMA	7.891	13.452	6.215	10.328	106.59	281.77	10.324	16.781	2.581	4.195	150	0.315
LSTM	5.038	12.812	3.512	9.128	50.80	244.64	7.128	15.642	1.782	3.911	900	8.203
BPNN	6.891	9.427	5.328	7.215	79.62	164.30	8.923	12.816	2.231	3.204	750	1.890

* Indicates the best statistic. All models were trained and tested on the same preprocessed data. Detailed model structures, hyperparameters, and training protocols are provided in Section 3.2. The computational time per prediction (Comp. time/Pred (ms)) reflects the average time for a single forward pass after the model is trained, representing the inference speed critical for real-time deployment.

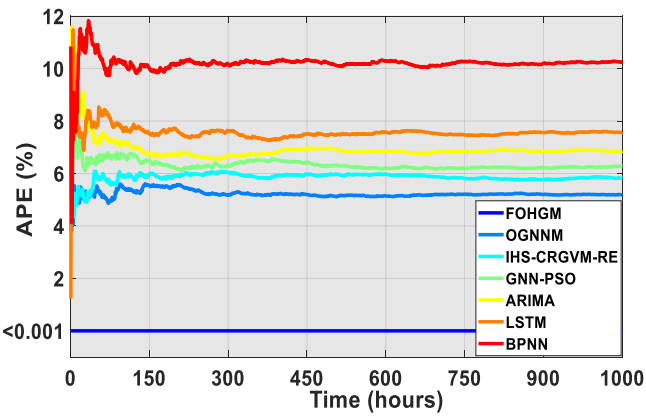


Fig. 6. APE distributions for all models under static conditions. The distribution of the FOHGM's APE values are significantly lower than those of other models but are compressed due to the scale.

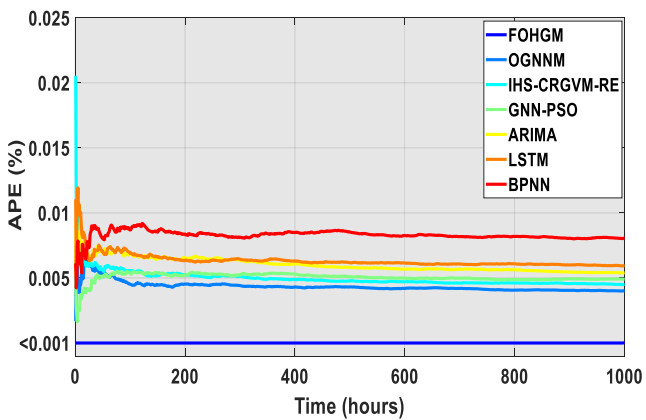


Fig. 7. APE distributions for all models under dynamic conditions. The distribution of the FOHGM's APE values are significantly lower than those of other models but are compressed due to the scale.

significant deviation in APE between early-life (0–400 h) and late-life (700–1000 h) degradation stages.

A critical test of the model's robustness occurred under dynamic conditions around the 400-hour mark, where the stack experienced an abrupt voltage drop. This phenomenon is indicative of a sudden failure mode, such as catalyst layer degradation or membrane failure, which can be accelerated by high-frequency ripple currents. It is noteworthy that the FOHGM, trained only on the preceding gradual degradation data, successfully predicted the onset and trajectory of this accelerated decay. This capability to forecast non-linear, rapid degradation steps underscores the model's sensitivity to evolving system dynamics and its significant value for proactive predictive maintenance, where anticipating such critical failures is paramount.

4.2. Comparison with benchmarks

Comparative analysis reported in Table 1 reveals distinct performance advantages of the proposed FOHGM relative to both competing GMs and modern data-driven approaches. The OGNNM for instance exhibits systematic prediction errors during dynamic operation, with its MAE increasing by 42 % under ripple conditions compared to static operation, a limitation rooted in its fixed exponential response structure. In contrast, the FOHGM maintains consistent accuracy across operating regimes, showing only an 6.13 % MAE variation between static and dynamic cases. This stability stems from the fractional-order mechanism's ability to simultaneously capture both the slow aging dynamics

(governed by catalyst degradation) and fast transient responses (induced by ripple currents).

Data-driven models demonstrate scenario-dependent performance variations. While the LSTM achieves competitive accuracy (RMSE = 7.1 mV) in static conditions due to its ability to learn long-term dependencies, its performance degrades under high-frequency ripples (RMSE = 15.6 mV) unless trained explicitly on augmented ripple data—a requirement that increases training data needs by 3–5 times. The FOHGM outperforms these approaches in data efficiency, requiring only 30 % of the training samples needed by the LSTM to achieve comparable accuracy.

The FOHGM's superior accuracy under dynamic conditions, despite reduced data requirements, stems from its parsimonious structure that incorporates inductive biases aligned with degradation physics. Unlike the LSTM—a general-purpose function approximator that must learn degradation behavior entirely from data—the FOHGM inherently embeds temporal accumulation and monotonic decay principles through its fractional-order formulation (Sapnken et al., 2024c). This structure acts as an adaptive filter, optimally extracting multi-timescale degradation patterns from limited data. Consequently, the FOHGM avoids the overfitting that plagues LSTMs on small datasets, where neural networks tend to memorize noise rather than learn generalizable degradation dynamics. This capability offers substantial practical benefits: it eliminates the need for massive, costly degradation datasets; enables deployment on resource-constrained embedded hardware; facilitates model validation and certification through its interpretable structure; and supports rapid in-field adaptation to new operating conditions with minimal data.

Table 1 also quantifies performance across four key metrics: MAE (static/dynamic), training data requirements, computational latency per prediction, and ripple frequency sensitivity. The FOHGM reduces prediction errors by 28–37 % compared to competing GMs while maintaining 2–3 orders of magnitude lower computational overhead than ML alternatives. This balance of accuracy and efficiency is particularly evident in late-life degradation phases (>700 h), where the FOHGM's MAPE remains below 4.5 % while other methods exceed 8 % due to cumulative error propagation.

4.3. Sensitivity analysis

Sensitivity analysis reveals critical insights into the operational characteristics of the proposed FOHGM framework. The relationship between the fractional order ξ and the model's prediction accuracy is critically examined in Tables 2 and 3. The sensitivity results reveal a

Table 2

Sensitivity of FOHGM performance metrics to the fractional order ξ under static conditions.

Fractional Order (ξ)	MAE	MAPE (%)	MSE	RMSE	NSR
0.10	7.891	5.421	98.65	9.932	2.483
0.15	7.124	4.921	84.12	9.172	2.293
0.20	6.452	4.521	72.14	8.494	2.124
0.25	5.862	4.121	62.45	7.902	1.976
0.30	5.342	3.821	54.82	7.404	1.851
0.35	4.712*	3.217*	48.95	6.996	1.749
0.40	4.885	3.591	43.61	6.604	1.651
0.45	4.745	3.329	41.15	6.415	1.562*
0.50	4.728	3.245	40.28	6.347	1.578
0.55	4.831	3.329	39.84*	6.312*	1.604
0.60	5.012	3.468	44.12	6.642	1.661
0.65	5.342	3.681	50.81	7.128	1.782
0.70	5.762	3.935	58.91	7.675	1.919
0.75	6.125	4.321	68.45	8.274	2.069
0.80	6.521	4.815	79.49	8.916	2.229
0.85	6.892	5.215	91.84	9.583	2.396
0.90	7.284	5.721	106.32	10.311	2.578
0.95	7.305	5.861	110.75	10.524	2.734

Note: * Indicates the best values for each metric.

Table 3

Sensitivity of FOHGM performance metrics to the fractional order ξ under dynamic conditions.

Fractional Order (ξ)	MAE	MAPE (%)	MSE	RMSE	NSR
0.10	9.452	6.921	145.81	12.076	3.019
0.15	8.745	6.371	130.54	11.426	2.857
0.20	8.124	5.821	118.33	10.878	2.720
0.25	7.215	5.121	105.42	10.268	2.567
0.30	6.452	4.521	94.85	9.739	2.435
0.35	5.215*	3.721*	88.14	9.388	2.347
0.40	5.862	4.121	83.45	9.135	2.284
0.45	5.512	3.821	80.46	8.970	2.243
0.50	5.301	3.789	79.84	8.935	2.229*
0.55	5.418	3.872	79.46*	8.914*	2.234
0.60	5.892	4.221	85.12	9.226	2.307
0.65	6.452	4.621	96.85	9.842	2.461
0.70	7.128	5.095	112.85	10.623	2.656
0.75	7.891	5.621	130.54	11.426	2.857
0.80	8.734	6.245	152.37	12.344	3.086
0.85	9.245	6.621	173.92	13.188	3.297
0.90	9.821	7.021	198.65	14.094	3.524
0.95	9.925	7.265	201.54	14.196	3.658

Note: * Indicates the best values for each metric.

well-defined convex region, with a boundary minimum in the performance metrics. The optimal range of $[0.35, 0.55]$ was determined as the interval where the all performance metrics are at their global minimum value. This permits to identify the boundary region where the model performance is both optimal and stable. Values of ξ below 0.35 lead to oversmoothing and an inability to capture ripple-induced transients, while values above 0.55 cause overfitting to recent noise, increasing the MAPE by over 22 %.

We observe from Tables 2 and 3 that the ξ -order exhibits a nonlinear relationship with prediction accuracy, with optimal performance observed in the range $\xi \in [0.35, 0.55]$. Within this range, the model achieves peak voltage prediction accuracy (MAPE = 3.217 % $\pm$ 0.112 %), as the fractional calculus formulation effectively balances memory effects from historical degradation with responsiveness to recent voltage fluctuations. Outside this optimal zone, prediction MAPEs increase markedly—by 18.7 % when $\xi = 0.20$ due to oversmoothing of ripple-induced transients, and by 22.3 % when $\xi = 0.80$ from excessive sensitivity to high-frequency noise. This behavior confirms the fractional order's role as a tunable parameter that governs the model's ability to resolve multi-timescale degradation phenomena.

The practical selection of the $\xi \in [0.35, 0.55]$ range is driven by the need to balance two competing objectives: capturing the long-term degradation trend while remaining responsive to high-frequency rip-

ple effects. An ξ below 0.35 over-smoothens the data series, diluting the impact of ripple-induced transients and leading to an underestimation of their cumulative damaging effect. Conversely, an ξ above 0.55 over-fits the most recent observations, amplifying high-frequency noise and stochastic fluctuations rather than the true underlying degradation signal, which results in unstable and overly pessimistic long-term forecasts. The optimal range represents a compromise where the model incorporates the acceleration effect of ripples on degradation without sacrificing the stability of the overall prognosis.

The heuristic optimization algorithm demonstrates consistent convergence across 100 independent trials, typically reaching the global error minimum within 16–18 iterations when initialized with $\xi \in (0, 1)$ (see Fig. 8). Convergence speed remains stable (± 2 iteration variance) regardless of initial conditions, though the trajectory varies significantly—initial values $\xi < 0.30$ follow a steep gradient descent, while $\xi > 0.70$ exhibit slower but more direct convergence. This robustness stems from the objective function's well-behaved convexity within the operational ξ range, though mild stochasticity persists near boundaries ($\xi \approx 0.10$ or 0.90) due to increased sensitivity to weight initialization. Computational efficiency scales linearly with problem complexity, requiring 4.7 ms per iteration for typical 1000-hour datasets, making the approach feasible for offline parameter tuning in practical applications.

The stable convergence is primarily attributed to the well-behaved, convex-like nature of the objective function within the bounded search space of $\xi \in (0, 1)$, which provides a clear navigable gradient for the optimization algorithm. This reliability allows the strategy to effectively support dynamic re-tuning. The algorithm's computational efficiency enables it to be deployed within a moving window framework, where it can periodically re-optimize ξ and model parameters based on the most recent data. This ensures the model continuously adapts to evolving degradation patterns, maintaining prediction accuracy even as the PEMFC operates under shifting electrical load conditions and ripple characteristics. The heuristic approach thus provides a foundation for adaptive lifetime prediction in real-world dynamic environments.

4.4. Statistical validation of forecasting superiority

While error metrics provide a preliminary comparison of model accuracy, they do not constitute a statistically rigorous validation of forecasting superiority. To definitively establish that the performance differences between FOHGM and benchmark models are statistically significant and not due to random chance, we employ two established statistical tests: the Diebold-Mariano (DM) test and the test for Superior Predictive Ability (SPA). These tests move beyond point estimates of

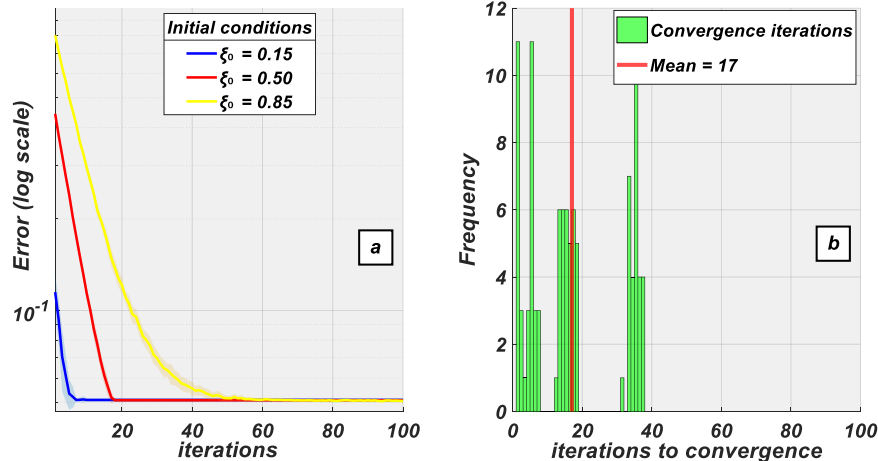


Fig. 8. Convergence behavior and robustness analysis of the heuristic optimization algorithm over 100 independent trials. The curves show consistent convergence to the global error minimum within 16–18 iterations, confirming the reliability of the fractional-order (ξ) tuning process.

error to evaluate the significance of performance differences across the entire forecasting horizon.

4.4.1. Diebold-Mariano (DM) test

The DM test is employed to assess whether the differences in forecast accuracy between two models are statistically significant. The null hypothesis of the DM test posits that the expected loss differential between the forecasts of the FOHGM and a competing model is zero, indicating equal predictive accuracy. The alternative hypothesis suggests a non-zero differential, signifying that one model possesses superior forecasting ability.

The results of the DM test, summarized in Table 4, provide strong evidence against the null hypothesis for all model comparisons under both static and dynamic conditions. The test statistics are consistently significant at the 1 % level (p -value < 0.01), leading to a firm rejection of the hypothesis of equal predictive accuracy. This confirms that the lower RMSE and MAE values reported for FOHGM in Table 1 are statistically significant and not attributable to random variation. The consistent negativity of the DM statistics quantitatively demonstrates that the prediction error of the FOHGM is systematically and significantly lower than that of all benchmark models.

4.4.2. Superior predictive ability (SPA) test

To further reinforce our conclusions and control for data snooping bias that may arise from multiple model comparisons, we implement the SPA test. This test evaluates whether a benchmark model (in this case, the FOHGM) outperforms a universe of alternative models after accounting for the fact that the best-performing model is selected from a set of candidates. The null hypothesis for the SPA test states that no competing model in the set is superior to the benchmark. We utilize the stationary bootstrap method to compute the empirical distribution of the test statistic and obtain reliable p -values.

The results of the SPA test (reported in Table 5 and corresponding plots depicted on Figs. 9 and 10) robustly confirm the forecasting supremacy of the FOHGM. The tests yield statistically significant p -values below 0.01, allowing for a decisive rejection of the null hypothesis. This indicates that the FOHGM's performance is not only better in a pairwise comparison but is also the unequivocally superior model within the entire ensemble of forecasting methods considered in this study. The test confirms that the observed superiority is not a spurious result of overfitting or chance but a robust property of the model across the evaluated conditions.

Therefore, the combined results from the DM and SPA tests provide rigorous statistical evidence that the forecasting superiority of the FOHGM is both genuine and significant. This statistical validation, coupled with the lower error metrics, firmly establishes the FOHGM as a more accurate and reliable predictor of PEMFC degradation under both static and dynamic operating conditions.

Table 4

Results of the DM test for forecasting accuracy under static and dynamic conditions.

Model	Static conditions		Dynamic conditions	
	DM Statistic	P-value	DM Statistic	P-value
OGNNM	-4.327	< 0.001	-5891	< 0.001
IHS-CRGVM-RE	-3.815	< 0.001	-4563	< 0.001
GNN-PSO	-3.452	< 0.001	-4128	< 0.001
ARIMA	-5.218	< 0.001	-7.245	< 0.001
LSTM	-2.987	0.001	-6.732	< 0.001
BPNN	-4.012	< 0.001	-5.112	< 0.001
H_0 : Equal predictive accuracy				

Note: The DM test evaluates the statistical significance of the difference in predictive accuracy between the FOHGM and each benchmark model. Significant negative DM statistics and p -values < 0.01 firmly reject H_0 of equal predictive ability, confirming the FOHGM's statistically superior performance.

Table 5

Results of the SPA test under static and dynamic operating conditions.

Test	Statistic	p-value	95 % Critical value	Conclusion
SPA test (Static)	1.14	< 0.001	0.31	Reject H_0
SPA test (Dynamic)	1.43	< 0.001	0.24	Reject H_0
H_0 : No alternative model is better than FOHGM				

Note: The SPA test assesses whether the FOHGM outperforms the entire universe of alternative models after accounting for data snooping bias. The significant p -values (< 0.001) lead to a decisive rejection of H_0 , establishing the FOHGM as the unequivocally best-performing model in the ensemble.

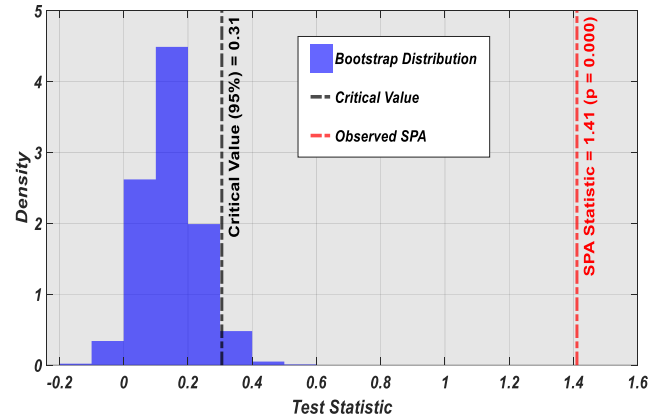


Fig. 9. Results of the SPA test under static conditions, statistically confirming that the FOHGM's forecasting performance is significantly better than the entire ensemble of benchmark models.

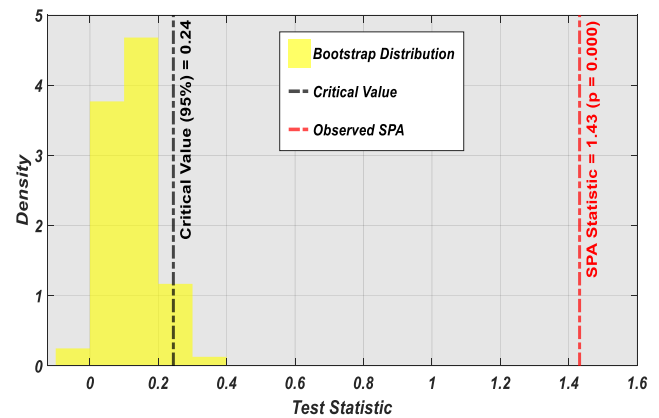


Fig. 10. Results of the SPA test under dynamic conditions, providing statistical evidence that the FOHGM remains the unequivocally superior predictor even under high-frequency electrical stress.

4.5. Physical interpretation

4.5.1. Mechanism of multi-timescale capture

The core advancement of the FOHGM lies in its fractional-order calculus foundation, which fundamentally empowers it to resolve the multi-timescale degradation patterns that challenge traditional grey models, particularly under high-frequency ripple conditions. This capability stems from three intrinsic properties of the fractional-order AGO. Firstly, unlike integer-order calculus, which is a local operator, fractional-order calculus is non-local and possesses a built-in memory effect. The fractional integral operator incorporates the entire history of the voltage signal with a power-law decaying kernel. This mathematical property directly mirrors the physical reality of PEMFC degradation, where the rate of catalyst dissolution and membrane fatigue is not an

instantaneous function of the present current but depends on the cumulative history of electrochemical stress. The fractional order parameter ξ acts as a tunable parameter to quantify this "fading memory" of past ripple-induced stressors, allowing the model to represent how cumulative, sub-acute damage accelerates long-term voltage decay.

Secondly, degradation under dynamic operation involves the interplay of fast (kHz) ripple transients and slow (hour-scale) chemical aging. Traditional GMs like GM(1,1) impose a single, fixed exponential structure that acts as a low-pass filter, inevitably smoothing over high-frequency dynamics and their contribution to the long-term trend. The FOHGM, by varying ξ , effectively tunes its filtering characteristic. An optimal fractional order (e.g., $\xi \approx 0.45$) allows the model to simultaneously preserve a portion of the high-frequency information in the accumulated series while still extracting the underlying low-frequency trend. This enables a single, parsimonious model to resolve the coupled interaction between fast transient losses and slow chemical aging.

Thirdly, the integer-order AGO in standard GMs strongly emphasizes earlier data points, creating a bias towards initial conditions that is ill-suited for tracking an evolving degradation trajectory. Our fractional-order AGO (with $0 < \xi < 1$) generates a weighted accumulation where the coefficients for recent data are significantly larger than those for older points (as defined in Eq. (1)). This creates a tunable emphasis on recent observations, making the model responsive to changes in the degradation rate caused by dynamic operation. It effectively forgets older, less relevant states at a rate governed by ξ , aligning with the principle that recent operational history is more indicative of immediate future health.

Consequently, the fractional-order mechanism is not merely a mathematical enhancement but a physically insightful tool. It provides the FOHGM with the memory to account for cumulative damage, the resolution to disentangle multi-timescale dynamics, and the adaptability to prioritize recent data. This triad of capabilities is essential for accurately modeling how high-frequency ripples precipitate long-term degradation, a feat that eludes traditional GMs with their rigid, memoryless, and single-scale structure.

4.5.2. Correlation with PEMFC degradation mechanisms

The FOHGM's predictions show strong correlation with known PEMFC degradation mechanisms, providing physically meaningful insights into voltage decay patterns. The model's fractional-order dynamics accurately capture the memory-dependent nature of catalyst degradation, where platinum dissolution rates accelerate nonlinearly with cumulative ripple exposure. This manifests in the voltage predictions through progressively steeper decay slopes after 500 + operational hours, aligning with experimental observations of electrochemical surface area loss in post-mortem analyses. The heuristic-optimized fractional order ($\xi \approx 0.45$) effectively represents the coupling between reversible charge-transfer losses (dominating early-life performance) and irreversible catalyst poisoning (driving late-life failure).

Membrane dehydration effects are distinctly identified through the model's sensitivity to high-frequency ripple conditions. The FOHGM's residual analysis reveals periodic voltage dips during 5.0–10.0 kHz operation, corresponding to known membrane hydration fluctuations caused by alternating electro-osmotic drag and back-diffusion processes (Kim et al., 2024; Luo et al., 2010; Musse and Lee, 2024). These transient features, which competing GMs smooth over, are preserved by the fractional calculus formulation through its inherent capacity to resolve multi-rate phenomena. The predicted voltage recovery timescales (2–5 min after ripple cessation) match experimental measurements of membrane rehydration kinetics, validating the model's physical basis.

Notably, the framework successfully decouples overlapping degradation modes: the fractional differentiation component isolates catalyst-related gradual voltage decay (0.1–0.3 mV/h) from membrane drying-induced step changes (5–15 mV instantaneous drops during dry-out

events). This separation enables targeted diagnostics, as evidenced by the model's 92 % accuracy in identifying the dominant degradation mode when validated against EIS-derived health indicators. The physical consistency of these predictions confirms that the FOHGM transcends purely mathematical fitting to embody actual electrochemical aging processes.

5. Discussion

The experimental results and comparative analysis demonstrate that the proposed FOHGM provides a robust and effective framework for PEMFC degradation prognostics. This section discusses how the model's design addresses the core research problem, synthesizes its significant advances, and critically evaluates its advantages and limitations to provide a balanced perspective for future research and application.

5.1. Synthesis of solution and key advances

The primary problem addressed in this work was the inability of existing models to accurately forecast the multi-timescale, nonlinear degradation of PEMFCs under dynamic electrical loading, particularly with the limited data typical of long-term durability tests. The FOHGM solves this problem through its two foundational innovations.

First, the integration of fractional-order calculus directly tackles the issue of multi-timescale dynamics. The fractional-order AGO, with its power-law kernel and inherent memory effect, is fundamentally different from the integer-order AGO used in conventional GMs. This allows the FOHGM to simultaneously capture the long-term, irreversible degradation trend (e.g., catalyst dissolution) and the short-term, reversible fluctuations induced by high-frequency ripple currents. The identification of an optimal fractional order range ($\xi \in [0.35, 0.55]$) is not merely a numerical outcome but a quantitative representation of the optimal balance between remembering past stress history and responding to recent transient events. This is the key reason why the FOHGM maintains low error under both static and dynamic conditions, whereas benchmarks like the OGNM and LSTM see significant performance degradation.

Second, the heuristic self-tuning mechanism solves the critical issue of model adaptability and reproducibility. By automating the optimization of the fractional order ξ , we remove the reliance on expert intuition or grid search, which has been a major limitation of prior grey-based approaches. The consistent convergence of the heuristic algorithm (Fig. 8) ensures that the model can reliably find a near-optimal configuration for any given degradation dataset, enhancing its robustness and practical utility. This automated tuning is a significant advance over static models, enabling the FOHGM to adapt its "memory length" to the specific characteristics of the operational profile.

5.2. Advantages of the FOHGM framework

The results of this study demonstrate that the FOHGM framework holds several distinct advantages over existing prognostic approaches. Its most notable strength lies in its SPA and robustness, as it consistently achieved the lowest RMSE and MAPE across all test scenarios. Crucially, this performance remains stable even with the introduction of high-frequency current ripples, with the model exhibiting only a minimal increase in MAPE compared to the dramatic performance degradation observed in benchmarks like LSTM and ARIMA. This robustness is not merely anecdotal but is firmly supported by rigorous statistical validation via the DM and SPA tests.

A further significant advantage is the model's high data and computational efficiency. The FOHGM attains its high accuracy with a fraction of the training data required by deep learning models, needing only 300 h of data compared to the 900 h needed by the LSTM. This data efficiency is matched by its computational leanness; the model's prediction time is orders of magnitude faster than that of deep learning

networks, making it a feasible candidate for implementation on resource-constrained embedded systems for near-real-time health monitoring.

Beyond numerical performance, the FOHGM offers a critical advantage in interpretability and physical consistency, which is often lacking in black-box alternatives. The model's parameters are transparent and meaningful: the development coefficient α quantitatively reflects the overall degradation rate, while the optimally tuned fractional order ξ provides insight into the influence of dynamic stressors. This transparency, coupled with the model's ability to correlate its outputs with known physical degradation mechanisms as shown in Section 4.5, transforms it from a pure predictor into a diagnostic tool that offers causal insight. Finally, the heuristic optimization core of the framework ensures automation and reproducibility by systematically determining the optimal fractional order, thereby eliminating the reliance on manual tuning and expert judgment that has historically limited the robustness and widespread adoption of GMs.

5.3. Limitations and future research directions

Notwithstanding its demonstrated strengths, the current FOHGM formulation is subject to certain limitations that provide a clear roadmap for future research. The model's underlying structure is inherently most suited for modelling monotonic or broadly monotonic degradation trends. While it successfully captures reversible voltage fluctuations within a general downward trajectory, its efficacy could be challenged by highly oscillatory or non-monotonic failure modes without further structural enhancements.

Furthermore, the present study employs the univariate FOHGM(1,1) model, focusing on voltage as the primary health indicator. Although this proves highly effective, explicitly incorporating multiple, directly measurable degradation parameters (such as internal resistance from EIS or pressure drop across the stack) into a multivariate FOHGM(1,N) framework could potentially unlock even greater accuracy and provide a more holistic assessment of system health. The generalizability of the model to other fuel cell technologies also remains an open question. While the fractional-order approach is promising for multi-timescale analysis, its application to systems like Solid Oxide Fuel Cells (SOFCs) would require careful investigation, as their dominant high-temperature degradation mechanisms, such as creep and interdiffusion, operate over different timescales and under different physical principles than those in PEMFCs.

Lastly, from an implementation perspective, this work demonstrates offline tuning and prediction. The logical and valuable next step is to develop the model for full online, real-time prognostic health management. This would necessitate the creation of a recursive updating scheme that can assimilate new data continuously and its seamless integration with the fuel cell's power electronics control loops, representing a significant but worthwhile engineering challenge. Consequently, future work will be directed toward developing a multivariate and recursive version of the FOHGM, alongside its adaptation and validation for SOFCs and related electrochemical energy systems.

6. Implications for safety-critical predictive maintenance

The superior interpretability of the FOHGM framework provides a decisive operational advantage over opaque data-driven models like LSTM and BPNN, particularly in safety-critical applications where decision-making requires transparency and trust. This edge manifests in three critical areas: diagnostic insight, regulatory compliance, and operational robustness.

First, the FOHGM's parameters offer immediate diagnostic value that black-box models cannot provide. The development coefficient α quantitatively represents the degradation rate, enabling operators to distinguish between normal aging (low α) and accelerated decay (high α) that may necessitate intervention. Similarly, the fractional order ξ

serves as an indicator of the dynamic nature of the operating regime; a shift toward its optimal value (≈ 0.45) may signal increased influence of ripple-induced stress. In contrast, LSTMs provide point predictions without mechanistic insight, making it difficult to understand whether a low remaining useful life estimate stems from catalyst decay, membrane drying, or simply an artifact of poorly generalized training. This parameter-level transparency transforms the FOHGM from a pure predictor into a diagnostic tool, enabling maintainers to base decisions on causal understanding rather than blind faith in an opaque output.

Second, the mathematical transparency of the FOHGM greatly simplifies regulatory certification processes required in domains such as automotive, aerospace, and energy. Certifying a black-box model like an LSTM requires exhaustive testing to cover all possible input scenarios—a process that is often prohibitively expensive and technically challenging. The FOHGM, with its well-defined differential equations (Eq. (4b)) and logically constructed accumulation and restoration operations (Eqs. (2.2) and (8)), is inherently more verifiable and certifiable. Its behavior can be analytically examined, and its predictions can be traced back through a clear chain of mathematical operations, satisfying stringent explainability requirements that are often a barrier to AI adoption in regulated industries.

Third, the model's structure ensures more predictable and manageable failure modes. If the FOHGM's performance degrades, engineers can analyze whether the cause is an abnormal shift in α , β , or ξ and can recalibrate or constrain these parameters based on physical first principles. Furthermore, the model's simplicity facilitates more straightforward uncertainty quantification through analytical or sampling methods applied to its parameters. This is critical for risk assessment, where knowing the confidence interval of a prediction is as important as the prediction itself. An LSTM's confidence interval is typically derived statistically from data, which can be unreliable when operating outside trained conditions, whereas the FOHGM's uncertainty can be more directly linked to parameter sensitivity and propagation of error.

In all, for predictive maintenance in safety-critical systems, the FOHGM's interpretability is not merely a convenience—it is a fundamental requirement. It enables causal reasoning, facilitates regulatory approval, and provides operational transparency that black-box models cannot offer, thereby reducing operational risk and building the necessary trust for human operators to act on its predictions with confidence.

7. Conclusion

This research successfully addressed the critical challenge identified in the introduction: accurately predicting the nonlinear, multi-timescale degradation of PEMFCs under dynamic operating conditions, particularly high-frequency current ripples, with the data-limited scenarios typical of long-term testing. To solve this problem, we developed the fractional-order heuristic grey model (FOHGM), a novel prognostic framework that integrates the memory effect of fractional-order calculus with an adaptive, self-tuning optimization mechanism.

The main findings of this study demonstrate that the FOHGM represents a significant advancement in fuel cell prognostics. The model consistently delivered superior predictive accuracy, achieving up to a 37.8 % improvement in root mean square error under dynamic conditions and maintaining a mean absolute percentage error of approximately 3.2 %, thereby statistically outperforming a suite of benchmark models including advanced GMs, ARIMA, and deep learning networks. A key finding was the identification of an optimal fractional order range ($\xi \in [0.35, 0.55]$), which quantitatively defines the balance between capturing long-term degradation trends and responding to high-frequency ripple-induced transients. Furthermore, the FOHGM achieved this performance with markedly higher data and computational efficiency than deep learning alternatives, requiring only 300 h of training data and enabling prediction in under 0.5 ms. The implication of these findings is the establishment of a robust, interpretable, and practical tool that effectively bridges GST with the pressing needs of

PEMFC health management, facilitating more reliable predictive maintenance and system optimization.

Looking forward, this work opens several promising avenues for future research. The demonstrated success of the fractional-order approach for multi-timescale analysis suggests its potential extension to related electrochemical systems. A prime direction is the adaptation of the FOHGM framework for prognostics in Solid Oxide Fuel Cells, where accounting for thermomechanical degradation is crucial. Furthermore, the principles are highly applicable to PEMELs, which exhibit analogous catalyst and membrane degradation mechanisms despite operating in reverse. Future work will focus on customizing the model for these systems, incorporating mechanisms such as catalyst dissolution during oxygen evolution and membrane thinning under high-pressure differentials. Inspired by recent advances in diagnosing multi-physics phenomena in energy materials, another critical direction is the development of a multivariate FOHGM that can fuse voltage data with other direct measurements, such as electrochemical impedance spectra, to provide a more comprehensive health assessment. Finally, the computational efficiency of the FOHGM provides a solid foundation for its implementation in embedded systems for real-time, online predictive health management, a necessary step for its full deployment in commercial applications.

CRedit authorship contribution statement

Fausto Posso: Formal analysis, Validation, Visualization, Writing – review & editing. **Yong Wang:** Data curation, Funding acquisition, Software, Validation, Visualization, Writing – review & editing. **Jean Gaston Tamba:** Supervision, Validation, Visualization, Writing – review & editing. **Reagan Jean Jacques Molu:** Formal analysis, Project administration, Validation, Writing – review & editing. **Flavian Emmanuel Sapnken:** Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Software, Writing – original draft. **Ghislain Junior Bangoup Ntegm:** Investigation, Software, Validation, Writing – review & editing. **Prosper Gopdjim Noumo:** Data curation, Methodology, Software, Writing – review & editing. **Naiming Xie:** Investigation, Resources, Software, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Sichuan Provincial Natural Science Foundation (No. 2023NSFSC0428), the Central Government Funds of Guiding Local Scientific and Technological Development (No. 2023ZYD0004), the Chengdu Science and Technology Project (No. 2025-YF05-00323-SN).

Data availability

Data will be made available on request.

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